

Palleti Vamshi

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[LinkedIn](#) • [GitHub](#) • [LeetCode](#) • [CodeChef](#) • [Codeforces](#)

PROFESSIONAL SUMMARY

AI/ML undergraduate at VNR VJiet (CGPA 9.9/10) with strong foundations in Data Structures and Algorithms. Experienced in engineering event-driven IoT telemetry pipelines, normalized relational databases, and decoupled backends from first principles.

EDUCATION

Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology (VNR VJiet) Hyderabad, India

B.Tech in Artificial Intelligence & Machine Learning • Expected Graduation: 2029 • **Current CGPA: 9.9/10**

Narayana Junior College — Intermediate • **987 marks**

Pratibha Model High School — Secondary School • **score 9.7**

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java (21 LTS), JavaScript (ES6+), SQL

AI & Machine Learning: Scikit-Learn, XGBoost, Machine Learning Foundations, Explainable AI (SHAP)

Backend Development: Spring Boot, Spring Data JPA, Node.js, Express.js, RESTful APIs

Databases & Systems: MySQL (3NF Relational Schemas), MongoDB (Mongoose), H2 In-Memory

Tools & Protocols: Git, GitHub, Apache Maven, Vite, MQTT (Mosquitto & Paho), Linux CLI

FEATURED TECHNICAL PROJECTS

LightX-IDS (Evolution of IDS_prototype)

[\[GitHub ->\]](#)

Technologies: Python, MQTT (Mosquitto), Scikit-Learn, XGBoost, Digital Twin Simulation

- Built an industrial digital twin simulating 6 physical machine classes streaming 10 real-time continuous sensor telemetry channels.
- Engineered an event-driven MQTT ingestion pipeline to validate, sanitize, and buffer sensor packets under bandwidth constraints.
- Trained and benchmarked tree-based classifiers (Decision Trees, Random Forest, XGBoost) optimized for edge inference and SHAP explainability.

Smart Campus Management System

[\[GitHub ->\]](#)

Technologies: Java 21, Spring Boot, Spring Data JPA, MySQL 8.x, H2 Database, Maven

- Architected a 16-table relational MySQL schema (3NF) governing departments, courses, faculty, attendance, and certificate requests.
- Implemented 16 Spring Data JPA repositories with Jakarta Validation, enforcing referential integrity and composite primary keys.
- Formulated 13 optimized DBMS queries for conflict-free timetable scheduling and validated services with automated in-memory H2 test suites.

TECHNICAL PROJECTS (CONTINUED)

Mentor-Student Management

[\[GitHub ->\]](#)

Technologies: Node.js, Express 5, MongoDB, Mongoose, JWT, BCrypt

- Developed an asynchronous REST service using Express 5 and ES modules for academic mentorship tracking during the P17 Hackathon.
- Designed MongoDB schemas with Mongoose 9, configuring index structures, relational references, and schema-level validation.
- Implemented stateless JWT authentication with BCrypt password hashing and secure HTTP cookie parsing middleware.

Attendance Management System

Technologies: Python, File I/O, Structured Data Processing, CSV Reporting

- Built a standalone Python utility to automate student attendance capture, roster validation, and aggregate summary metrics.
- Implemented duplicate-entry validation routines and structured CSV export workflows for administrative record-keeping.

HONORS & COMPETITIVE ACHIEVEMENTS

Code Frenzy — 22nd rank

Competitive programming contest ranking.

LEADERSHIP & INVOLVEMENT

AWS Student Builder Group — Volunteer in Tech & Innovation

Engaged in technical exploration initiatives, community knowledge sharing, and emerging technology sessions.

COMPETITIVE PROGRAMMING & CODING PROFILES

LeetCode (@vamsh_i2007): Active problem solving covering binary search, dynamic programming, trees, and graphs.

CodeChef (@vamsh_i2007): Regular participant in timed competitive programming rounds and algorithmic contests.

Codeforces (@vamsh_i2007): Contest participation with rigorous asymptotic runtime and memory constraints.

GitHub (@palleti-vamshi): Public codebases, industrial IoT simulation prototypes, and academic project implementations.

CORE ENGINEERING STRENGTHS

Data Structures & Algorithms • Relational Schema Design • Systems Architecture • Critical Thinking • Technical Curiosity